

TotallyRealEstate — Relational Schema

CLIENT (

client_id PK,

first_name,

last_name,

email ,

phone,

created_at

)

AGENT (

agent_id PK,

license_number ,

first_name,

last_name,

email ,

phone,

is_active

)

STAFF (

staff_id PK,

first_name,

last_name,

email,

role_code,

is_active

)

PROPERTY (

property_id PK,

property_ref ,

address_text,

municipality,
property_type,
bedrooms ,
bathrooms ,
floor_area_sqm

)

LISTING (

listing_id PK,
property_id FK → PROPERTY.property_id,
agent_id FK → AGENT.agent_id,
asking_price,
listed_on,
closed_on,
status

)

LISTING_SELLER (

listing_id PK/FK → LISTING.listing_id,
client_id PK/FK → CLIENT.client_id

)

SHOWING (

showing_id PK,
listing_id FK → LISTING.listing_id,
agent_id FK → AGENT.agent_id,
starts_at,
ends_at,
status

)

SHOWING_BOOKING (

booking_id PK,
showing_id FK → SHOWING.showing_id,

```
    client_id FK → CLIENT.client_id,  
    party_size,  
    status,  
    booked_at  
)  
  
OFFER (  
    offer_id PK,  
    listing_id FK → LISTING.listing_id,  
    submitted_by_agent_id FK → AGENT.agent_id,  
    offer_amount,  
    submitted_at,  
    expires_at ,  
    accepted_at ,  
    status
```

```
OFFER_BUYER (  
    offer_id PK/FK → OFFER.offer_id,  
    client_id PK/FK → CLIENT.client_id  
)
```

```
SALE_TRANSACTION (  
    sale_id PK,  
    offer_id FK → OFFER.offer_id,  
    coordinator_staff_id FK → STAFF.staff_id,  
    completed_on,  
    sale_price  
)
```

Foreign key	Referenced key	Parent → child participation	On delete	On update
LISTING.property_id	PROPERTY.property_id	1 → 0..N	RESTRICT	RESTRICT
LISTING.agent_id	AGENT.agent_id	1 → 0..N	RESTRICT	RESTRICT
LISTING_SELLER.listing_id	LISTING.listing_id	1 → 0..N; 1..N when listing is non-DRAFT	RESTRICT	RESTRICT
LISTING_SELLER.client_id	CLIENT.client_id	1 → 0..N	RESTRICT	RESTRICT
SHOWING.listing_id	LISTING.listing_id	1 → 0..N	RESTRICT	RESTRICT
SHOWING.agent_id	AGENT.agent_id	1 → 0..N	RESTRICT	RESTRICT
SHOWING_BOOKING.showing_id	SHOWING.showing_id	1 → 0..N history; at most 3 active	RESTRICT	RESTRICT
SHOWING_BOOKING.client_id	CLIENT.client_id	1 → 0..N	RESTRICT	RESTRICT
OFFER.listing_id	LISTING.listing_id	1 → 0..N	RESTRICT	RESTRICT
OFFER.submitted_by_agent_id	AGENT.agent_id	1 → 0..N	RESTRICT	RESTRICT
OFFER_BUYER.offer_id	OFFER.offer_id	1 → 1..N	RESTRICT	RESTRICT
OFFER_BUYER.client_id	CLIENT.client_id	1 → 0..N	RESTRICT	RESTRICT
SALE_TRANSACTION.offer_id	OFFER.offer_id	1 → 0..1	RESTRICT	RESTRICT
SALE_TRANSACTION.coordinator_staff_id	STAFF.staff_id	1 → 0..N	RESTRICT	RESTRICT

Functional dependencies

Table	Main functional dependency	Other key/dependency notes
CLIENT	client_id → first_name, last_name, email, phone, created_at	Nullable email is unique when present, but is not a candidate key for the whole table.
AGENT	agent_id → license_number, first_name,	license_number and email are

Table	Main functional dependency	Other key/dependency notes
	last_name, email, phone, is_active	additional candidate keys and each determines all agent attributes.
STAFF	staff_id → first_name, last_name, email, role_code, is_active	email is an additional candidate key and determines all staff attributes.
PROPERTY	property_id → property_ref, address_text, municipality, property_type, bedrooms, bathrooms, floor_area_sqm	property_ref is an additional candidate key; address_text is not a declared key.
LISTING	listing_id → property_id, agent_id, asking_price, listed_on, closed_on, status	property_id is not a key because listing history is retained. agent_id does not determine property/listing details.
LISTING_SELLER	listing_id + client_id → no non-key attributes	All-key relation; neither participating ID alone uniquely identifies a row.
SHOWING	showing_id → listing_id, agent_id, starts_at, ends_at, status	The responsible agent may differ from the listing agent. Listing ID does not determine session times or showing agent.
SHOWING_BOOKING	booking_id → showing_id, client_id, party_size, status, booked_at	The active pair is conditionally unique, not a candidate key for the full history.
OFFER	offer_id → listing_id, submitted_by_agent_id, offer_amount, submitted_at, expires_at, accepted_at, status	The submitting agent may differ from the listing agent. listing_id is unique only within accepted offers.
OFFER_BUYER	offer_id + client_id → no non-key attributes	All-key relation; neither participating ID alone uniquely identifies a row.
SALE_TRANSACTION	sale_id → offer_id, coordinator_staff_id, completed_on, sale_price	offer_id is an additional candidate key and determines every sale attribute.

NORMALIZATION STEPS

Final table	1NF → 2NF outcome	2NF → 3NF outcome
CLIENT	Extract client-only facts from listing/seller and offer/buyer composite rows	Reuse for lead-client facts in bookings; no further non-key dependency is assumed
AGENT	Agent details can initially remain in the single-key listing/offer/session working relations	Extract facts determined by the agent identifier; reuse the one AGENT table for all three roles
STAFF	Flat sale relation already has single-column candidate keys	Extract coordinator details determined by coordinator_staff_id
PROPERTY	Property details can initially remain with the single-key listing working relation	Extract facts determined by property_id; property_ref is an alternate key
LISTING	Extract listing-only facts from the (listing_id, client_id) flat relation	Remove property/agent details; keep their FKs and listing-specific values
LISTING_SELLER	Retain the seller participation pair after removing partial dependencies	All-key relation; no further split
SHOWING	Booking working relation needs no partial-key split	Extract session facts from bookings, then retain agent_id instead of copied agent details
SHOWING_BOOKING	booking_id is a single-column key; no 2NF split required	Remove session/client details; retain facts about the particular party booking
OFFER	Extract offer-only facts from the (offer_id, client_id) flat relation	Remove submitting-agent and any copied listing/property details; retain FKs and offer facts
OFFER_BUYER	Retain the buyer participation pair after removing partial dependencies	All-key relation; no further split
SALE_TRANSACTION	sale_id and offer_id are single-column candidate keys; no 2NF split required	Remove coordinator details; retain sale facts and references